

WORKSHOP & HANDS-ON LAB

k6 in the Age of AI: Copilot Skills & MCP-k6

Augmenting modern performance engineering with curated AI skills and Model Context Protocol.

— Almudena Vivanco · Grafana Champion

The Evolution of Performance Testing

Legacy Testing

Manual scripts and isolated test runs.

Developer-Centric k6

Code-driven testing embedded directly in CI/CD.

AI-Assisted Engineering

Curated agentic skills driving test generation.

CLAUDE CODE v2.1.224

CROSS-SESSION MESSAGING

\$ claude --name developer

TWO SESSIONS. ONE BUILDS. ONE TESTS. ■

one PRD, two named sessions,
zero copy-paste between windows.
no Xcode window, terminal only.

@21zerixpm

JRDEVHUB.COM

● ● ● developer

- > reading prd-clean.md
- > screen 3 of 7 built
- > msg -> QA

STEP 1 DONE 2 BUGS FOUND

● ● ● QA

- \$ npx serve-sim
- > iPhone 16 Pro booted
- > req 2.4 FAILED

Core Architecture of k6



JS Scripting

Native ES6 test logic and modular helpers.



Virtual Users

Parallel goroutine execution with low memory overhead.



Strict Thresholds

Pass-fail criteria for automated CI pipelines.

The Pitfall of Raw LLM Generation

80%

Of unguided AI tests contain silent flaws

Hallucinated endpoints, weak status assertions, and missing workload thresholds.

Introducing copilot-k6-skills

Guiding Copilot with domain rules transforms generic code into production-grade k6 scripts.

- ✓ **Domain Rules:** Enforce idiomatic k6 metrics.
- ✓ **Assertion Safety:** Validate response schemas strictly.
- ✓ **Standard Options:** Predefined ramp-up stages.

```
// .github/copilot-instructions.md
export const options = {
  stages: [{ duration: '30s', target: 50 }],
  thresholds: {
    http_req_duration: ['p(95)<250'],
  },
};
```

Components of a k6 Copilot Skill



Context Rules

Project endpoints, auth flows, and headers.



Load Profiles

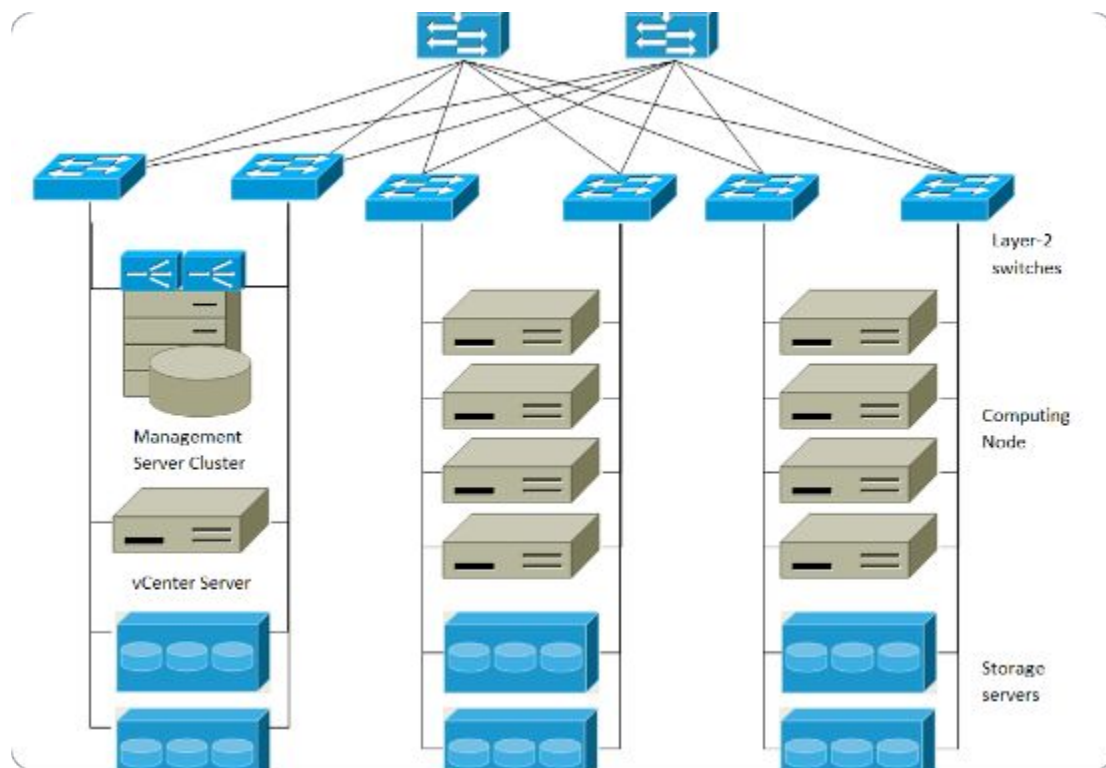
Standard smoke, stress, and soak patterns.



Validation Sets

Mandatory status checks and response assertions.

Connecting k6 via Model Context Protocol



Standardized MCP Interface

Connects AI agents directly to k6 CLI execution.

Iterative Test Generation

Natural language prompting runs real load tests.

Real-time Telemetry Feedback

Metrics stream straight back into the agent loop.

The Agentic Testing Loop

01

Prompt

Target endpoint & user goals.

02

Synthesize

Skills shape test structure.

03

Execute

MCP runs local k6 engine.

04

Review

Human engineer validates SLAs.

Hands-on Lab Milestones

Lab 1

Environment & Basic Scripting

Install k6 OSS and author initial load scripts.

Lab 2

Configuring Copilot Skills

Load repository skills into GitHub Copilot.

Lab 3

Agent Execution via MCP-k6

Run automated conversational performance iterations.

Lab 4

Failure Triage & Reporting

Diagnose threshold breaches and review metrics.

Raw Prompting vs. Skill-Guided k6

Raw AI Prompts

- ✗ **Brittle assertions:** Misses status verification.
- ✗ **Unrealistic loads:** Flat concurrency spikes.
- ✗ **No SLAs:** Missing threshold parameters.

Skill-Guided k6

- ✓ **Rigorous checks:** Validates body and headers.
- ✓ **Realistic profiles:** Accurate ramp-up curves.
- ✓ **Enforced SLAs:** Built-in latency percentiles.

Workshop Productivity Outcomes

3x

Faster test authoring

100%

Threshold coverage

0

Silent hallucination errors

GET STARTED TODAY

Build Your k6 AI Skills

Access instructions, skills templates, and sample configurations on GitHub.

github.com/MrsDaehin/copilot-k6-skills

Questions · Hands-on Troubleshooting · Open Discussion

Image Sources



https://miro.medium.com/1*2uXvAYpVSnD7uU9lxhpHLg.png

Source: 21zerixpm.medium.com



https://docs.cloudstack.apache.org/en/4.19.3.0/_images/large-scale-redundant-setup.png

Source: docs.cloudstack.apache.org